AET weekend implementation roadmap

Prepared for David Parla on 16 October 2026 Implementation weekend: Saturday 17 and Sunday 18 October 2026 All schedule times: Europe/Madrid, CEST

Decision and scope

Build a complete, valid residual-shape pipeline on Saturday. On Sunday, add curve-conditioned historical memory and run at most one more experimental method. Protect the final two hours for reproducibility, documentation and packaging.

Recommended starting stack: Python, NumPy, pandas, SciPy, scikit-learn, LightGBM and pytest. The proposed core methods run on CPU; measure runtime on your machine. This plan assumes approximately 16 focused engineering hours and familiarity with Python. It is a plan to implement, not an already implemented model.

The strongest additional idea is to retrieve historical residual shapes using the hourly curve of the day being reconstructed. The most interesting mathematical experiment is a constrained point predictor built from conditional quantiles. Cross-zone graph shrinkage is a cheaper alternative experiment.

These are adaptations of established methods to AET, not claims of original research. Only the challenge brief was provided; no official training CSV was available for empirical testing. No AET accuracy gains or runtimes are claimed. A small artificial distribution was used only to check the quantile derivation, never to train a model or choose its hyperparameters.

Nonnegotiable challenge boundaries

The official training period is 1 October 2025 through 30 September 2026, with DST transition days excluded: 363 days, 139392 quarter-hour rows and 34848 zone-hour targets. The confirmed evaluation date is 19 October 2026; other dates may also be scored. [1]

During development this weekend, do not consult or use 1–18 October 2026 prices, even if public. Research papers and standard code libraries may inform the method; external datasets, pretrained weights, external embeddings and holiday calendars must not enter the solution. Use the supplied anonymized zones without attempting to identify them.

After the freeze, jury-supplied earlier history may feed rules already fixed in code. Do not refit models, update fitted scalers, change neighbor selection rules, tune blend weights or recalibrate from that history. Recomputing a declared lag or fixed weighted summary is feature calculation.

For every hour, output four finite prices whose average differs from the supplied hourly price by no more than 0.0001 EUR/MWh. MAE is averaged by day, then by zone, then equally across the four zones. The brief gives no numerical weighting between qualitative and quantitative assessment.

Sunday 18:00 below is a personal freeze target. The brief does not establish that as the official submission deadline.

Research priorities

Effort estimates below are incremental engineering time after the data and backtest framework exists. They are planning estimates.

Method	Why it is worth investigating	Timebox	Priority
Cumulative-curve interpolation	Preserves hourly averages algebraically and gives a useful shape baseline	30–45 min	Core
Historical residual summaries	Supplies recent shape information under a frozen model	45–60 min	Core
Curve-conditioned analogue memory	Selects relevant history using the known current-day hourly curve	2 h	Best additional experiment
Alternative Haar basis	Tests whether half-hour structure is easier to learn than slope and zigzag	30 min	Cheap ablation
Cross-zone graph shrinkage	Shares shape information more when supplied hourly curves agree	45–60 min	Optional
Constrained quantile point prediction	Allocates the mean correction according to conditional uncertainty	2 h prototype	Optional and experimental

A Cumulative interpolation as a conservative baseline

Temporal disaggregation methods explicitly reconstruct fine-resolution values while preserving coarse aggregates. Denton and related approaches provide useful context; they are established methods, not new discoveries. Their smoothing assumptions should be tested against AET's sawtooth patterns. [2]

Use an especially small implementation for the weekend. For one zone, index hourly prices H[0:24], and define cumulative boundaries:

```
C[0] = 0
C[h+1] = C[h] + H[h]
F = an interpolator through (0, C[0]), ..., (24, C[24])
P_base[h,q] = 4 * (F(h + (q+1)/4) - F(h + q/4))
             for h = 0,...,23 and q = 0,...,3
```

The four differences telescope, so their mean is H[h]. Compare this with flat prices and a centered linear ramp. Start with SciPy PCHIP; its shape-preserving behavior applies to the cumulative curve, not a guarantee that the resulting prices are optimal or free of artifacts. Signed and negative hourly prices remain valid inputs. [6]

If a cumulative spline performs poorly, keep the linear ramp. Do not add a separate reconciliation stage to an already feasible basis output and expect it to create information. Do not enforce continuity between consecutive market blocks.

B Curve-conditioned analogue memory

Weighted nearest-neighbor methods have been studied for electricity price forecasting. The AET adaptation is to retrieve a fine-resolution residual shape using an hourly curve that is already known, rather than predict an unknown price level. The cited work establishes precedent for weighted neighbors, not evidence of a gain on this challenge. [3]

Build this as a deterministic historical feature extractor:

1. Derive each historical day's hourly means and baseline from its own 15-minute observations.
2. Store the zero-sum residual r = observed price - baseline price.

3. For a target zone and hour, consider only earlier dates at the same zone and hour. Begin with a 180-calendar-day search window.
4. Describe each candidate and query using local hourly differences at offsets -2 to +2, daily level and range, current-hour cross-zone spreads, and weekday/weekend. Use one-sided features and missing flags at day boundaries.
5. Standardize continuous descriptors with a scaler fitted only on the model's training slice. Keep that scaler frozen when more history arrives.
6. Retrieve K neighbors using descriptor distance. Start with K in {8, 16, 32}; keep other choices fixed initially.
7. Average their four residuals with one normalized weight per neighbor, shared across all four quarters. A fixed recency factor may be added only as a separate ablation.
8. Supply the retrieved residual coefficients, nearest-neighbor distance, sample count and effective neighbor count to the residual learner.

```
weights[j] >= 0
sum_j weights[j] = 1
memory[q] = sum_j weights[j] * historical_residual[j,q]
effective_count = 1 / sum_j weights[j]**2
```

The retrieved residual remains zero-sum. Sparse or missing history returns zeros plus an explicit missing flag. Do not combine a spline with a full historical deviation without checking for duplicated ramp information.

To avoid retrospective leakage when training, compute every row's memory from dates before that row's delivery date. Never retrieve the same day's other quarters or another zone's same-day 15-minute targets.

Inference appends no fitted parameters: it performs a fixed lookup and summary over the supplied history. Freeze window length, distance definition, scaler, K, weighting and fallback before submission. Log selected historical dates for a few diagnostic examples.

For the quick baseline before retrieval, use previous-day and previous-week residual coefficients plus 7- and 28-day means. Arithmetic means preserve zero sum; coordinate-wise medians generally do not.

C A half-hour representation as a cheap alternative

Your original basis is complete and orthogonal. A different full basis cannot add information, but can change what separately trained or regularized learners find easy:

```
half-hour contrast = [ 1,  1, -1, -1] / 2
first-half detail  = [ 1, -1,  0,  0] / sqrt(2)
second-half detail = [ 0,  0,  1, -1] / sqrt(2)
```

These three vectors are orthonormal and zero-sum. This is a proposed ablation based on the algebra of the task, not a claimed market fact. Compare the original normalized basis and this Haar basis under the same features, splits and model budget. Also compare retaining only the strongest predictable components.

Do not judge coefficient MAE. Reconstruct all four prices and score those prices.

D Cross-zone graph shrinkage

Research on multi-market electricity forecasting has used graph models to exploit inter-market dependence. A weekend-sized adaptation is a four-node shrinkage operator, without training a neural network. The published results concern a different forecasting task and do not establish benefits here. [4]

For each hour, let C_hat be a 4-by-3 matrix of predicted total deviation coefficients, one row per zone. Define symmetric similarities from supplied hourly prices:

```
A[z,k] = exp(-abs(H[z] - H[k]) / s), for z != k
A[z,z] = 0
L = diag(A.sum(axis=1)) - A
C_smoothed = solve(I + lambda * L, C_hat)
```

Choose positive scale s from training-only spread statistics with a fixed floor. Test lambda in {0, 0.1, 0.3}; lambda = 0 is the incumbent. All choices freeze before inference. Restore each zone's own H after decoding its coefficients.

Every zone still has a zero-sum shape, so hourly means stay intact. The graph expresses price similarity, not geographic connectivity or observed power flows. Equal hourly averages do not imply equal quarter-hour shapes: discard this method if it oversmooths a zone or only wins in one fold.

E Constrained point prediction from conditional quantiles

This is the more speculative option. A 2026 working paper studies reconciliation optimized for quantile loss and explains why squared-loss reconciliation does not generally solve that problem. Crucially, its coherent forecast distributions can still yield point quantiles that violate aggregation constraints. Its full algorithm therefore is not a drop-in AET solution. [5]

A smaller AET-specific derivation is possible. Let R_q be the baseline residual for quarter q, conditioned on all allowed features X, including the known hourly curve. Solve:

```
minimize over a[1:4]: sum_q E[abs(R_q - a_q) | X]
subject to:          sum_q a_q = 0
```

For continuous conditional marginal distributions F_q, the first-order conditions are:

```
2 * F_q(a_q) - 1 + multiplier = 0
therefore a_q = Q_q(tau) for a common tau
choose tau such that sum_q Q_q(tau) = 0
```

This is a constrained point decision. It is not a claim that marginal medians, quantiles or a full joint distribution must themselves add up. The separable expected absolute loss only requires the conditional marginals; the optimization does not require an independence assumption.

Prototype recipe:

1. Stack four quarter rows per hour with quarter_index as a feature.
2. Train five pooled LightGBM quantile models at levels {0.1, 0.3, 0.5, 0.7, 0.9}, targeting residuals in EUR/MWh. LightGBM supports quantile loss and the alpha parameter. [7]
3. Sort the five predicted quantile values within each quarter to remove crossings, then interpolate monotonically between levels.
4. Find a common tau by bisection when the summed quantile curves bracket zero.
5. If there is no bracket or any invalid value, use the four predicted medians followed by centering; retain the baseline/flat fallback chain.
6. Apply a final numerical mean correction and evaluate reconstructed prices.

Under known continuous marginals this solves the stated constrained expected-L1 decision. With a learned, finite quantile grid it is an approximation; model error and crossing repair can erase any benefit. Record bracket-failure rate and post-correction MAE.

I checked the derivation against independent numerical optimization on an artificial coherent distribution. That confirms the algebra only. It is not AET validation and supplies no tuning information.

Choose this OR graph shrinkage for Sunday's optional two-hour slot. Do not attempt both if the core pipeline is unfinished.

Core model and implementation contracts

Use the original normalized zero-sum basis for the first implementation. If B has orthonormal columns and r is the four-dimensional baseline residual, the training coefficients are B.T @ r.

A practical first learned model is three small pooled LightGBM regressors with market_id as a categorical feature and objective=regression_l1. This is a convenient surrogate: coefficient L1 differs from price L1. Select every configuration and the final number of trees using reconstructed-price MAE on earlier validation

data. Do not early-stop on the locked September block.

A joint regularized linear model with loss sum(abs(r - B @ W @ x)) is an interpretable alternative if you already have a convex solver ready. It is not necessary to add a new solver during the weekend.

Start with num_leaves=15, min_child_samples=100, learning_rate=0.05, a fixed seed and a small tree-count grid such as {200, 500}. These are initial search settings, not claimed optima. Keep all quarter and zone observations from a date in the same temporal partition.

Initial features: own hourly level; +/-1 and +/-2 hourly differences; curvature; hour and weekday; weekend; cyclic season; day range and negative-hour count; cross-zone spreads and slopes; lagged residual summaries. Daily hourly input from every zone is permitted. No division by H: use raw EUR/MWh or a positive floored scale learned from training.

Module	Responsibility
schema.py	Input columns, uniqueness, complete hours and quarters, finite values
basis.py	Encode and decode; original and optional Haar basis
baselines.py	Flat, linear ramp and cumulative interpolation
history.py	Strict earlier-date filtering, calendar lags, summaries and analogue memory
features.py	Same feature construction in training and inference
models.py	ShapeModel interface, fitted residual models and fixed optional postprocessors
backtest.py	Frozen cutoffs, history replay, official metric and experiment records
train.py	Fit selected configuration and save all necessary artifacts
infer.py	Read jury inputs, reconstruct, validate, write and reread output
tests/	Numerical invariants, history exclusion and offline end-to-end smoke checks

Keep this as a small package, not a framework. Use array shapes [day, zone, hour, quarter] internally where that makes grouping explicit.

Suggested commands to implement, not existing commands:

```
python -m aet.backtest --train data/train_prices_15min.csv --config configs/core.yaml
python -m aet.train \
  --train data/train_prices_15min.csv \
  --config configs/final.yaml \
  --out artifacts
python -m aet.infer \
  --hourly test_hourly.csv \
  --history history_15min.csv \
  --model artifacts \
  --out predictions.csv
```

Store model files, feature order, basis, frozen transforms, history rules, blend weights, dependency versions and seed in artifacts. Keep a manifest with the training CSV hash and actual training date range. Inference must not need the original training CSV or network access; persist any necessary training-history fallback explicitly.

Validation protocol

Use three development folds and one locked final check. For each evaluation date, replay all permitted history strictly before that date, including earlier dates after the model cutoff. Keep models and fitted transforms frozen for the whole evaluation block.

Role	Model training ends	Scored delivery dates	Extra history available initially
Development fold 1	31 March 2026	19–30 April 2026	1–18 April
Development fold 2	31 May 2026	19–30 June 2026	1–18 June
Development fold 3	31 July 2026	19–31 August 2026	1–18 August
Locked final check	31 August 2026	19–30 September 2026	1–18 September

The 18-day gap approximates the known training-end/evaluation-start separation while testing multiple seasons. It does not imply that only that horizon matters. If time remains, add a short near-cutoff diagnostic without introducing a new model search.

October 2025 is useful for descriptive shape analysis, but it has almost no earlier official training history. It cannot provide a mature-model forward validation comparable to October 2026.

For each run record: overall official MAE, four zone MAEs, each fold MAE, per-quarter error, daily paired difference versus the incumbent, maximum hourly-mean error, fallback count, train time and inference time. Retain per-day predictions for diagnosis. The effective evidence is repeated delivery dates, not 139392 independent samples.

Suggested promotion gate for extra complexity: at least 0.5% lower development MAE, improvement on at least two of three folds, and no zone more than 2% worse. These are pragmatic predeclared engineering thresholds, not significance claims. If results are close, keep the simpler method. Inspect daily paired differences to check that a single unusual day is not driving the result.

Freeze the selected method before opening the locked block. Use that block to report the generalization check, not to start another tuning cycle. After selection, fit the frozen configuration on all official training dates through 30 September.

Saturday 17 October

Time	Work	Concrete exit condition
09:00–10:00	Create package, schema checks, grouping and flat inference	A held-out hourly input produces a compliant CSV
10:00–11:30	Implement temporal replay and official scoring	Three cutoff configurations; date assertions; saved flat scores
11:30–13:00	Add linear and cumulative baselines; basis transforms	One baseline comparison and mean preservation verified after CSV reload
13:00–14:00	Lunch	Stop working
14:00–16:00	Build features and the small pooled residual model	First learned candidate scored on all development folds
16:00–17:00	Add causal residual history and test cross-zone features	Compare without history, with history, and with cross-zone additions
17:00–18:00	Run a complete train/infer cycle; draft methodology	A minimum viable submission exists, even if the winning model is simple

Saturday scope control: if backtesting is not trustworthy by 13:00, stop adding models and finish it. If a learned model does not beat the baseline, retain the baseline and diagnose target alignment, history leakage and basis reconstruction.

Sunday 18 October

Time	Work	Concrete exit condition
09:00–11:00	Implement curve-conditioned analogue memory	Same folds, causal neighbors, fixed descriptor transform; compare to simple summaries
11:00–12:00	Inspect errors and run the cheap basis/shrinkage ablation	A named incumbent and a short evidence table
12:00–13:00	Optional experiment, first half	Graph shrinkage OR constrained quantiles; no unrelated search
13:00–14:00	Lunch	Stop working
14:00–15:00	Finish optional experiment and decide	Promote only if the gate is met; otherwise remove from final inference
15:00–16:00	Freeze configuration, run locked check, train on all official data	Final artifacts and a recorded September check
16:00–18:00	Verify offline execution, document and package	Reproducible submission; output reloaded and validated; personal freeze at 18:00

If you have only eight hours total: spend one on schema/flat output, two on validation and baselines, two on a residual learner, one on historical summaries, and two on packaging. Drop analogue retrieval and all stretch methods.

Experiment ledger and stopping rules

ID	Candidate	Question
E0	Flat	What is the official comparison score?
E1	Linear ramp and cumulative interpolation	How much shape is recoverable without learning?
E2	E1 plus residual learner	Is the residual predictable from current-day inputs?
E3	E2 plus causal historical residual summaries	Does fresh permitted history help a frozen model?
E4	E3 plus analogue-memory features	Does matching the current hourly curve improve on fixed averages?
E5	Alternative basis or fixed shrinkage	Is a simpler representation or less amplitude more robust?
E6	One optional graph or quantile method	Does a more specialized assumption earn its cost?

Ablate one change at a time. Cap broad model configurations at roughly a dozen and save all results, including losses. Use one global blend first; add per-zone blend weights only if evidence justifies the extra parameters.

A convex blend of valid predictions remains valid when each model's weight is shared across the four quarters and weights sum to one. A coordinate-wise median ensemble does not automatically preserve the mean.

Do not add a transformer, diffusion model, online optimizer or large hyperparameter sweep this weekend. Foundation models and external embeddings are prohibited. High-frequency realism is useful only when the chosen shape improves the actual score or another explicitly evaluated requirement.

Definition of done

☐ Only permitted inputs influence training, selection and inference.
☐ Current-day hourly curves from all zones are allowed; current-day quarter-hour targets never enter features.
☐ Historical feature generation is causal during both training and backtesting.
☐ A sentinel future history row cannot change an earlier day's prediction.
☐ Calendar lags respect excluded DST dates; row offsets are not mistaken for dates.
☐ Missing or short history has documented behavior and never requires refitting.
☐ Every output key is unique and every required hour has exactly quarters 1–4.
☐ Prices are finite; negative values are preserved when predicted.
☐ Float64 final correction and sufficient CSV precision retain error <= 0.0001 EUR/MWh after reload.
☐ Learned output falls back to baseline, then flat, for valid finite hourly input.
☐ Repeated inference is deterministic within the documented numeric tolerance.
☐ Training and inference have separate documented entry points.
☐ A clean offline inference smoke run succeeds with saved artifacts.
☐ Dependencies are pinned; any required environment or wheel bundle is documented.
☐ Methodology includes the winning ablations, rejected ideas, data boundaries, metric, runtime and limitations.
☐ The actual organizer deadline and packaging requirements have been checked.

Evidence and reading order

Read [1] first for compliance, [2] for the interpolation foundation, and [7] for implementation. The other papers provide research context. Nothing in these sources substitutes for experiments on the official dataset.

[1] AET, Neural Wave 2026 challenge brief, supplied PDF, sections 5–12. Authoritative source for the data window, freeze, 0.0001 tolerance, evaluation and deliverables.

[2] Christoph Sax and Peter Steiner, Temporal Disaggregation of Time Series, The R Journal, 2013.

https://journal.r-project.org/articles/RJ-2013-028/

Establishes aggregate-preserving temporal disaggregation and the roles of Denton and regression-based alternatives. The cumulative-interpolation recipe above is our proposed lightweight implementation.

[3] Alicia Troncoso and coauthors, Electricity Market Price Forecasting Based on Weighted Nearest Neighbors Techniques, IEEE Transactions on Power Systems, 2007.

https://idus.us.es/items/ac140d42-8745-4345-8dfa-0cdf52ec0f0a

DOI: 10.1109/TPWRS.2007.901670 The university repository's indexed abstract establishes the weighted-neighbor precedent. Full-text performance results were not reviewed or transferred to AET.

[4] Yi Li and coauthors, Multi-task Graph Adaptive Learning for Multivariate Electricity Price Short-Term Forecasting in Australia's National Electricity Market, IEEE Transactions on Power Systems, 2025.

https://scholars.cityu.edu.hk/en/publications/multi-task-graph-adaptive-learning-for-multivariate-electricity-p/

DOI: 10.1109/TPWRS.2024.3380834 Author-affiliated abstract supports inter-market graph modeling as a research direction. Our four-node shrinkage rule is a different, untested adaptation.

[5] Nam Ho-Nguyen, Hossein Alipour, Anastasios Panagiotelis and George Athanasopoulos, Optimal Forecast Reconciliation for Quantiles, Monash working paper 04/26, April 2026.

https://www.monash.edu/business/ebs/research/publications/ebs/2026/wp04-2026.pdf

Supports aligning reconciliation with the relevant loss and explicitly distinguishes coherent distributions from additive point quantiles. Our constrained common-quantile decision is separately derived; it is not their algorithm.

[6] SciPy, PchipInterpolator documentation.

https://docs.scipy.org/doc/scipy/reference/generated/scipy.interpolate.PchipInterpolator.html

Implementation reference for shape-preserving cumulative interpolation. Pin the actual tested SciPy version.

[7] LightGBM, Parameters documentation.

https://lightgbm.readthedocs.io/en/stable/Parameters.html

Implementation reference for regression_l1, quantile and alpha. Pin the actual tested LightGBM version.

Research accessed on 16 October 2026. No October 2026 electricity-price data was consulted.